Detailed Notes: NLP Summarization, Translation and Question Answering

1. Types of Summarization
Summarization is the process of reducing a long text into a shorter version while preserving the important information.

Extractive Summarization:
• Selects important sentences or phrases directly from the original text.
• No new words are generated.
• Easier to implement and computationally efficient.
• Example: Selecting key sentences from a news article.

Abstractive Summarization:
• Generates completely new sentences that capture the meaning of the original text.
• Similar to how humans summarize text.
• Uses deep learning and transformer models.
• Example: ChatGPT summarizing an article in its own words.

Applications: News summarization, document compression, report generation, meeting summaries.

2. Translation in NLP
Translation converts text from one language to another.
Features:
• Supports multilingual communication.
• Uses transformer-based architectures such as Transformer, BERT, T5, and Gemini.
• Improves global accessibility.
• Used in Google Translate, virtual assistants and multilingual chatbots.

Modules in Translation Systems:
1. Language Conversion – converts source language to target language.
2. Advanced Models – transformer-based neural networks improve accuracy.
3. Multilingual Support – enables communication across multiple languages.
4. Global Accessibility – removes language barriers worldwide.
5. Real-world Applications – customer support, education, travel, healthcare.

3. Question Answering Systems (QA Systems)
QA systems are AI applications that understand user questions and provide relevant answers.
Working Process:
1. Understand the user's intent.
2. Retrieve relevant information from a knowledge base or external sources.
3. Generate an accurate response.
4. Present the answer conversationally.

Features:
• Natural language understanding.
• Information retrieval.
• Response generation.
• Conversational interaction.

Applications: Chatbots, virtual assistants, search engines, customer support systems.

4. Summarization Workflow Example
Step 1: Store the long article in a string variable.

Step 2: Split text into words using article.split().
Step 3: Count total words using len(words).
Step 4: Send the text to Gemini or another LLM using generate_content() for summarization.
Output: A concise summary preserving important information.

Python Example:
article = "Artificial Intelligence is transforming healthcare, education, finance and transportation."
words = article.split()
print(len(words))

5. Translation Workflow Example
Step 1: Define the sentence: sentence = "Good Morning"
Step 2: Split the sentence into words.
Step 3: Use a dictionary-based translator:
dictionary = {"Good":"■■■■■", "Morning":"■■■■"}
Step 4: Modern NLP systems use LLMs and transformer models for accurate contextual translation.

6. Question Answering Workflow Example
Step 1: Create a knowledge base.
Example:
context = "India's capital is New Delhi. India became independent in 1947."
Step 2: User asks a question.
Step 3: NLP model retrieves relevant information.
Step 4: Model generates and returns the final answer.

Important Concepts for Exams:
• Difference between Extractive and Abstractive Summarization.
• Role of Transformer models in NLP.
• Working of Translation Systems.
• Components of Question Answering Systems.
• Real-world applications of NLP summarization and translation.